Set-up

Index (1) is MSFT, (2) is AAPL. Market-cap weights and dollar positions, with $V_t = \$1{,}000{,}000$:

$$w_i = \frac{cap_i}{cap_1 + cap_2}, \quad w = \left(\frac{448.77}{1023.88}, \frac{575.11}{1023.88}\right) =$$

$$(0.438303, 0.561697),$$

$$\theta_t = S_t \lambda_t = V_t w = (438{,}303.32,\ 561{,}696.68).$$

To keep the two thetas apart, the EWMA parameter is written $\vartheta = 0.97$ below and θ_t is always the dollar position. Write $\mu = \mu_{t+\Delta}$, $\Sigma = \Sigma_{t+\Delta}$, $\sigma_i = \sqrt{\Sigma_{ii}}$, $\rho = \Sigma_{12}/(\sigma_1\sigma_2)$, $z = \Phi^{-1}(0.95) = 1.644854$.

Step 2 - VaR of the linearized loss and the square-root-of-time rule

$L^{lin}_{t+\Delta} = -\theta_t^T X_{t+\Delta}$ is a linear function of a normal vector, so

$$L^{lin}_{t+\Delta} \mid \mathcal{F}_t \sim N(m, s^2), \quad m = -\theta_t^T \mu, \quad s^2 = \theta_t^T \Sigma \theta_t.$$

$P(L \le l) = \Phi((l-m)/s)$ is continuous and strictly increasing, so the α-quantile solves $\Phi((l-m)/s) = \alpha$:

$$\boxed{\mathrm{VaR}_\alpha(L^{\mathrm{lin}}_{t+\Delta}) = -\theta_t^T \mu + \sqrt{\theta_t^T \Sigma \theta_t}\, \Phi^{-1}(\alpha)}$$

With $m = -1{,}270.29$, $s = 9{,}412.54$: $\mathrm{VaR}_{0.95} = -1{,}270.29 + 1.644854 \times 9{,}412.54 = 14{,}211.95$.

Square-root-of-time rule.

If the K daily losses $-\theta_t^T X_{t+k\Delta}$ were i.i.d. $N(0, s^2)$, their sum would be $N(0, Ks^2)$ and its VaR would be $\sqrt{K}\, s z = \sqrt{K}\, \mathrm{VaR}_\alpha(L^{\mathrm{lin}}_{t+\Delta})$. The rule applies this scaling to the one-day VaR as computed:

$$\mathrm{VaR}_\alpha(L^{\mathrm{lin}}_{t+K\Delta}) \approx \sqrt{10} \times 14{,}211.95 = 44{,}942.15, \quad 3 \times \mathrm{VaR} = 134{,}826.44.$$

Step 3(ii) – The conditional distribution and the shock in closed form

Define $\varepsilon = X_{(1)} - \mu_{(1)} - \beta(X_{(2)} - \mu_{(2)})$ with $\beta = \rho\sigma_1/\sigma_2$. Then $\mathbb{E}\varepsilon = 0$ and

$$\mathrm{Cov}(\varepsilon, X_{(2)}) = \rho\sigma_1\sigma_2 - \beta\sigma_2^2 = 0.$$

$(\varepsilon, X_{(2)})$ is jointly normal (a linear map of X), so zero covariance means independence, and

$$\mathrm{Var}(\varepsilon) = \sigma_1^2 - 2\beta\rho\sigma_1\sigma_2 + \beta^2\sigma_2^2 = \sigma_1^2 - \rho^2\sigma_1^2 = \sigma_1^2(1-\rho^2).$$

Hence, given $X_{(2)} = x_{(2)}$, $X_{(1)} = \mu_{(1)} + \beta(x_{(2)} - \mu_{(2)}) + \varepsilon \sim N(\tilde{\mu}_{(1)}, \tilde{\sigma}^2_{(1)})$ with the $\tilde{\mu}_{(1)}$, $\tilde{\sigma}_{(1)}$ of the sheet.

The shock.

Put $x_{(2)} - \mu_{(2)} = -5\sigma_2$ into $\tilde{\mu}_{(1)}$ and subtract $5\tilde{\sigma}_{(1)}$:

$$x_{(1)} - \mu_{(1)} = \frac{\rho\sigma_1}{\sigma_2}(-5\sigma_2) - 5\sigma_1\sqrt{1-\rho^2} = -5\sigma_1 c, \quad c := \rho + \sqrt{1-\rho^2}.$$

With $\rho = 0.239902$: $c = 1.210699$, so MSFT moves $5c = 6.05$ of its own standard deviations.

$$x_{(2)} = 0.001166 - 5(0.012331) = -0.060488,$$

$$x_{(1)} = 0.001405 - 5(0.011237)(1.210699) = -0.066619.$$

Loss on the shock day.

$$-\theta_t^{\top}x = 5(\theta_{t,1}\sigma_1 c + \theta_{t,2}\sigma_2) - \theta_t^{\top}\mu = 5(5{,}963.0 + 6{,}926.3) - 1{,}270.3 =$$

$$63{,}175.0.$$

Step 3(iii) - EWMA after the shock

The deviation from the old mean is $d = x - \mu = -5(\sigma_1 c, \sigma_2)^{\top}$, so

$$\mu_{t+2\Delta} = \lambda\mu + (1-\lambda)x = \mu + (1-\lambda)d = \mu - 0.15(\sigma_1 c, \sigma_2)^{\top}$$

$$= (-0.000636,\ -0.000684)^{\mathsf{T}},$$

$$\Sigma_{t+2\Delta} = \vartheta\Sigma + (1-\vartheta)\,dd^{\mathsf{T}} = \vartheta\Sigma + 25(1-\vartheta)\begin{pmatrix} \sigma_1^2 c^2 & \sigma_1\sigma_2 c \\ \sigma_1\sigma_2 c & \sigma_2^2 \end{pmatrix}.$$

Reading off the entries, with $25(1-\vartheta) = 0.75$:

$$\sigma_1'^2 = \sigma_1^2(\vartheta + 25(1-\vartheta)c^2), \qquad \sigma_2'^2 = \sigma_2^2(\vartheta + 25(1-\vartheta)),$$

$$\rho' = \frac{\vartheta\rho + 25(1-\vartheta)c}{\sqrt{(\vartheta + 25(1-\vartheta)c^2)(\vartheta + 25(1-\vartheta))}}.$$

None of this depends on μ. Numerically $\vartheta + 0.75c^2 = 2.0694$ and $\vartheta + 0.75 = 1.72$:

$$\sigma_1' = 1.4385\,\sigma_1 = 0.016165, \qquad \sigma_2' = 1.3115\,\sigma_2 = 0.016171,$$

$$\rho' = \frac{0.2327 + 0.9080}{\sqrt{2.0694 \times 1.72}} = \frac{1.1407}{1.8866} = 0.6046.$$